

Experiment No: 01

Experiment Name: Implementation of straight through cable connection.

Objective:

1. To study about different types of network cables.
2. To prepare the straight through cable connection using crimping tool.

Materials:

1. RJ-45 Connectors.
2. CAT-5 Cable.
3. Crimping tool.
4. Network cable tester.

Theory:

Cable is a medium through which information usually moves from one network device to another. There are several types of cable which are commonly used with LANs. In some cases, a network will utilize only one type of cable, other networks will use a variety of cable types. The types of cable chosen for a network is related to the network's topology, protocol and size. Understanding the characteristics of different types of cable and how they related to other aspect or a network is necessary for the development of a successful network.

Several kinds of cables:

1. Twisted Pair Cable:
 - Unshielded Twisted Pair (UTP) Cable.
 - Shielded Twisted Pair (STP) Cable.
2. Coaxial Cable etc.

Connecting cable:

Connecting a network cable is a very skilled task. There are generally two types of connection:

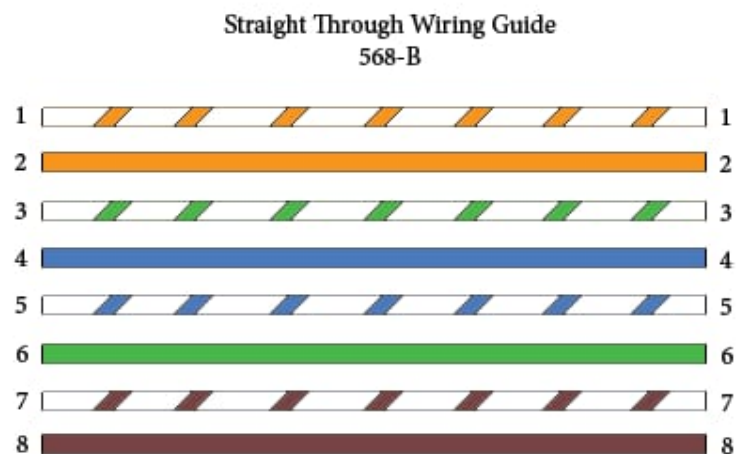
1. Straight Through Connection
2. Crossover Connection

As our aim is to implement straight through connection, we will briefly discuss about it and study about the implementation process.

Straight Through Connection:

A straight through cable is a type of twisted pair cable that is used in local area networks to connect a computer to a network hub such as a router. This type of cable is also sometimes called a patch cable and is an alternative to wireless connections where one or more computers access a router through a wireless signal.

Diagram of Straight Through Connection:



Procedure:

To crimp an ethernet cable the following steps are followed:

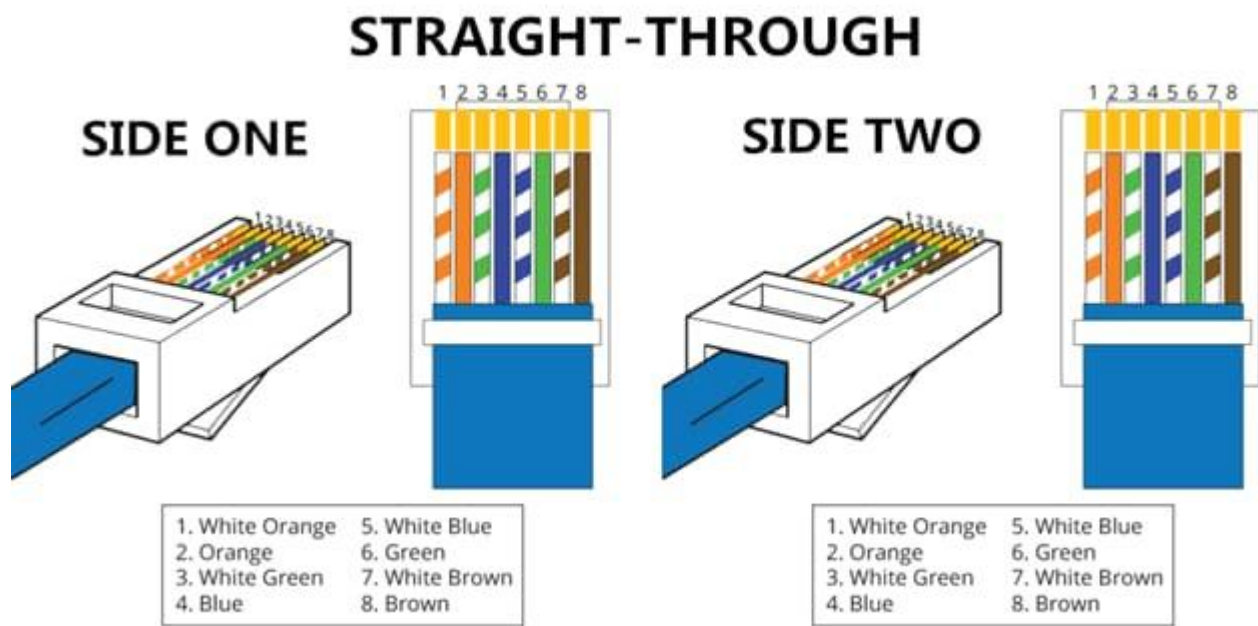
- To strip the cable and organize the cables wire, here we have taken CAT-5 ethernet cable. So, after stripping we will find and 8 colored wires inside the ethernet cable. There we have to put them into the correct ordering of colors.
- Untwisting the wires we have to straighter out wires. After that we have to put them into the correct order. In this step we will make sure that the wires are all flat and in line with each other.
- The ordering of the cables are following:

Side one

Side two

Striped orange	←	→	Striped orange
Solid orange	←	→	Solid orange
Striped green	←	→	Striped green
Solid blue	←	→	Solid blue
Striped blue	←	→	Striped blue
Solid green	←	→	Solid green
Striped brown	←	→	Striped brown
Solid brown	←	→	Solid brown

The figure will be:



- Trimmed the wires evenly to about ½ inch in length with the blade of our crimping tool. We wanted to make sure that we have enough room for the wires to reach the end of the RJ-45 connector. We also tried to have room for the shielding of the cable to be inserted into the connector too.
- In this step we will crimp it. Pushed the RJ-45 connector into the slot of our crimping tool for RJ-45 connectors. The slot should be labeled something like "8P" for the 8 pin RJ-45 connector that we are using.
- In this step we are doing the actual crimping part and compressing stabilizing the 8 golden pins on the RJ-45 connector into the 8 colored wires

Application:

In networking, straight-through cables are frequently utilized for the following purposes:

- **Connecting Computers to Hubs or Switches:** This method connects endpoints, such as PCs, to hubs or network switches so that they may communicate.
- **Connecting Routers to Modems:** Enables data transfer for internet access between a modem and a router.
- **Linking Network Devices to Access Points:** This method is used to integrate networks by connecting devices, like desktops and laptops, to wireless access points.
- **Printer or Server Connections:** Provides shared access by connecting printers or servers to the network through switches.
- **VoIP Phone Connection:** This method is used to link IP phones to switches so that voice calls can be made via a network.

The main purpose of straight-through cables is to link various kinds of devices together within a network.

Caution:

- Ensure proper wiring sequence on both ends.
- Keep cable length within 100 meters to avoid signal loss.
- Crimp connectors securely to prevent loose connections.
- Test the cable with a tester after crimping.
- Avoid using straight-through cables where crossover cables are needed.

Conclusion:

In this lab, we explored the fundamental purpose and functionality of straight-through cable connections, which are primarily used to connect devices of different types, such as computers to switches or routers. We also learned the detailed process of creating a straight-through Ethernet cable by correctly arranging the wires according to the T568A or T568B standard and securely crimping the RJ45 connectors. This hands-on experience helped us understand both the theoretical and practical aspects of setting up reliable network connections.

Experiment No: 02

Experiment Name: Implementation of cross-wired cable connection.

Objective:

1. To study about different types of network cables.
2. To prepare the cross-wired cable connection using crimping tool.

Materials:

1. RJ-45 Connectors.
2. CAT-5 Cable.
3. Crimping tool.
4. Network cable tester.

Theory:

Cable is a medium through which information usually moves from one network device to another. There are several types of cable which are commonly used with LANs. In some cases, a network will utilize only one type of cable, other networks will use a variety of cable types. The types of cable chosen for a network is related to the network's topology, protocol and size. Understanding the characteristics of different types of cable and how they related to other aspect or a network is necessary for the development of a successful network.

Several kinds of cables:

1. Twisted Pair Cable:
 - Unshielded Twisted Pair (UTP) Cable.
 - Shielded Twisted Pair (STP) Cable.
2. Coaxial Cable etc.

Connecting cable:

Connecting a network cable is a very skilled task. There are generally two types of connection:

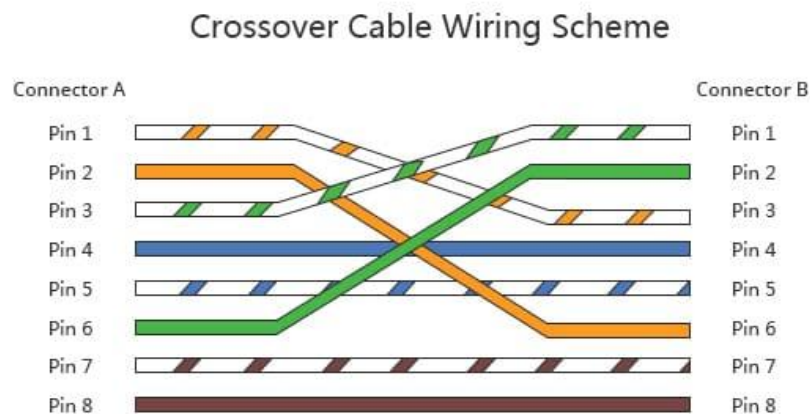
1. Straight Through Connection
2. Crossover Connection

As our aim is to implement cross-wired connection, we will briefly discuss about it and study about the implementation process.

cross-wired Connection:

A cross-wired cable is a type of cable installation that is used for the interconnection of two similar devices. It is enabled by reversing the transmission and receiving pins at both ends, so that output from one computer becomes input to others, and vice versa.

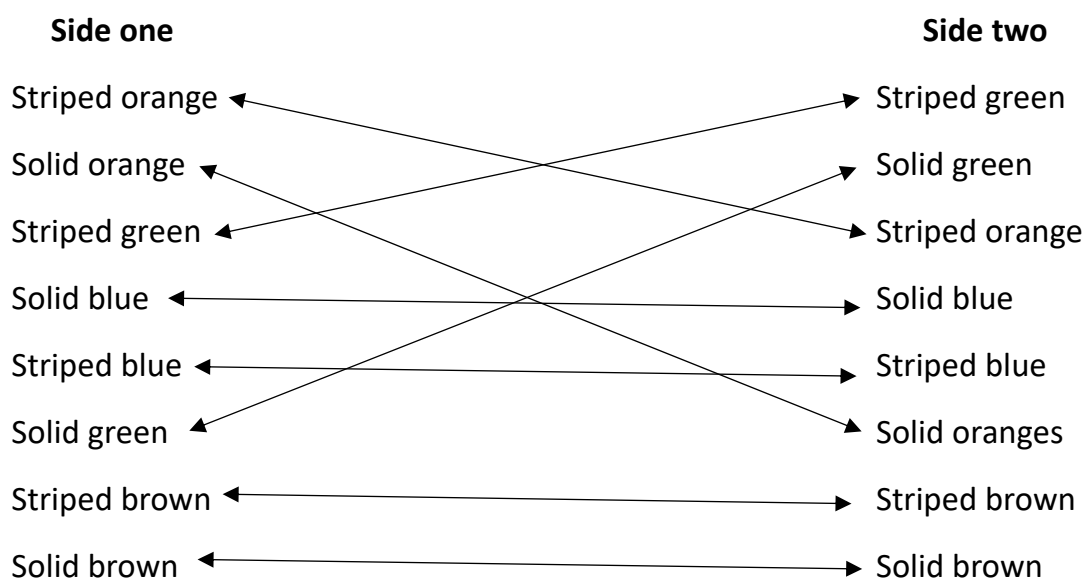
Diagram of Straight Through Connection:



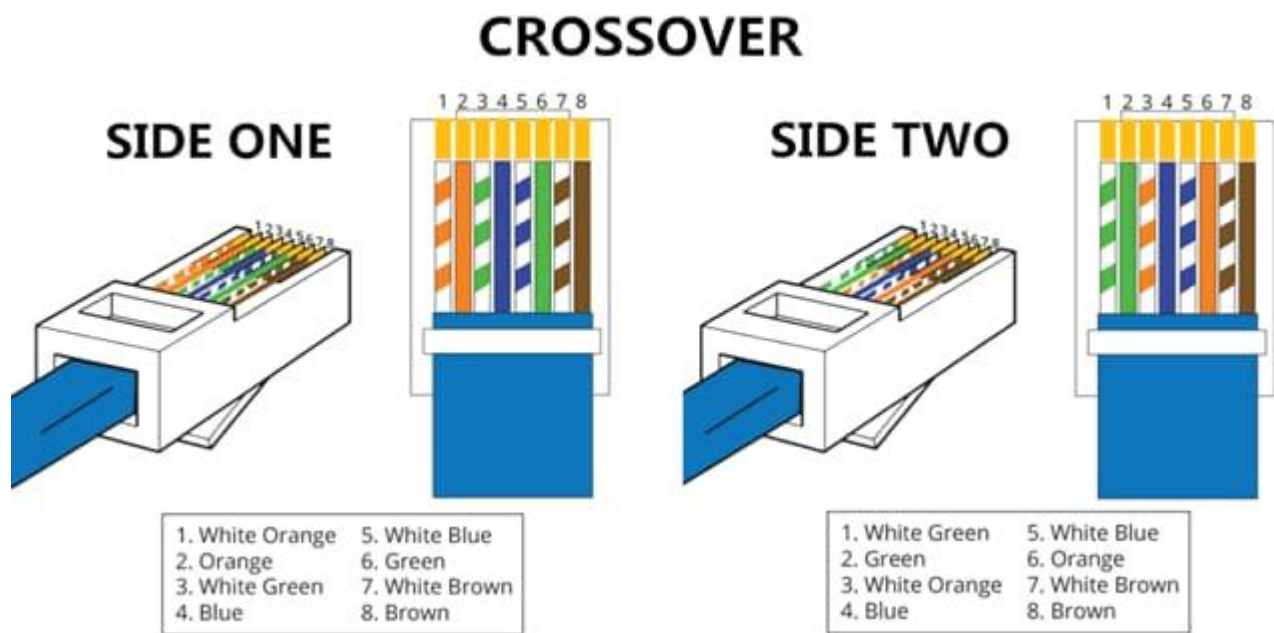
Procedure:

To crimp an ethernet cable the following steps are followed:

- To strip the cable and organize the cables wire, here we have taken CAT-5 ethernet cable. So, after stripping we will find and 8 colored wires inside the ethernet cable. There we have to put them into the correct ordering of colors.
- Untwisting the wires we have to straighter out wires. After that we have to put them into the correct order. In this step we will make sure that the wires are all flat and in line with each other.
- The ordering of the cables are following:



The figure will be:



- Trimmed the wires evenly to about ½ inch in length with the blade of our crimping tool. We wanted to make sure that we have enough room for the wires to reach the end of the RJ-45 connector. We also tried to have room for the shielding of the cable to be inserted into the connector too.
- In this step we will crimp it. Pushed the RJ-45 connector into the slot of our crimping tool for RJ-45 connectors. The slot should be labeled something like "8P" for the 8 pin RJ-45 connector that we are using.
- In this step we are doing the actual crimping part and compressing stabilizing the 8 golden pins on the RJ-45 connector into the 8 colored wires.

Application:

In networking, cross-wired cables are frequently utilized for the following purposes:

- Direct PC-to-PC Connection: Enables file sharing or communication between two computers without needing a switch or hub.
- Switch-to-Switch Communication: Used to link two switches that don't support auto-MDI/MDIX for network expansion.
- Router-to-Router Connection: Facilitates communication between two routers for advanced networking setups.
- Hub-to-Hub Linking: Connects hubs directly when auto-MDI/MDIX is not available.
- Testing in Labs: Commonly used for experiments or troubleshooting direct device-to-device connections.

The main purpose of cross-wired cables is to link various kinds of devices together within a network.

Caution:

- Ensure proper wiring sequence on both ends.
- Keep cable length within 100 meters to avoid signal loss.
- Crimp connectors securely to prevent loose connections.
- Test the cable with a tester after crimping.
- Avoid using straight-through cables where crossover cables are needed.

Conclusion:

In this lab, we explored the fundamental purpose and functionality of straight-through cable connections, which are primarily used to connect devices of different types, such as computers to switches or routers. We also learned the detailed process of creating a straight-through Ethernet cable by correctly arranging the wires according to the T568A or T568B standard and securely crimping the RJ45 connectors. This hands-on experience helped us understand both the theoretical and practical aspects of setting up reliable network connections.

Experiment No: 03

Experiment Name: Building a peer-to-peer network and sharing resources.

Objective:

1. Design and build a simple peer-to-peer network using a crossover cable.
2. Verify connectivity between the peers sharing resources.

Resources Required:

1. Two Windows 10 PCs, each with functional Network Interface Card (NIC).
2. An Ethernet crossover cable.

Procedure:

Step 1: Diagram the network.

(a) A network diagram is a map of the logical topology of the network. In the space below, we will sketch a simple peer-to-peer network connecting two PCs. Label one PC with IP address 192.168.1.1 and the other PC with IP address 192.168.1.2



(b) A simple network like the one we designed can use a hub or switch as a central connecting device, or the PCs directly connected. In this case, we will connect the PCs directly.

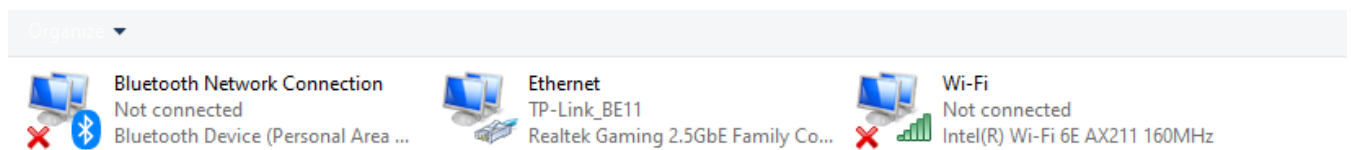
Step 2: Document the PCs.

(a) Checking the computer name settings for each PC and making adjustments as necessary. For each PC, select we have to select Start and Control Panel. Then we will double-click the system then click the Change settings on Computer name section. Here we will write down the computer name that is displayed following Full Computer Name.

PC ₁ Name	DESKTOP-4E K9HDB9
PC ₂ Name	LAPTOP-GRJS6SAM

(b) Checking to see that if the two PCs have the same name. If they do, we have to change the name of one PC by clicking the change button, typing a new name in the Computer name field, then we will click OK.

(c) Then we will click ok to close the System Properties window.



Step 3: Connect the Ethernet Cable.

(a) As per roll we use the Ethernet crossover cable provided by the instructor. We have to plug one end of the cable into the Ethernet NIC of PC1.

(b) Then we will plug the other end of the cable into the Ethernet NIC of PC2. As we insert the cable, we will hear a click which indicates that the cable connector is properly inserted into the port.

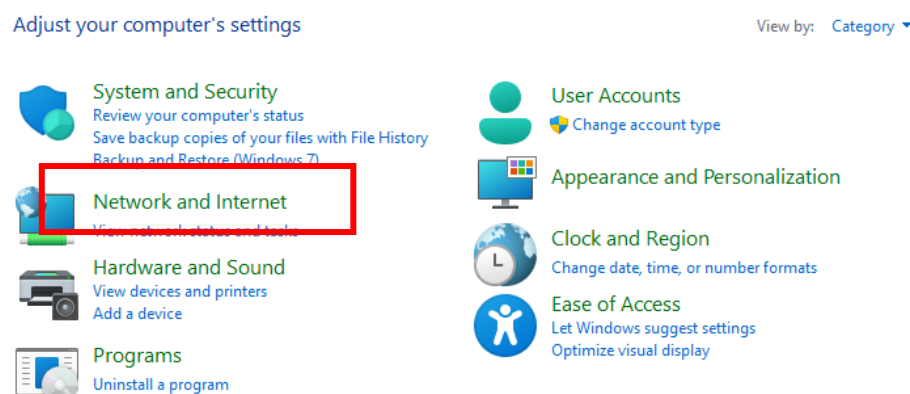
Step 4: Verify physical Connectivity.

(a) After the Ethernet crossover cable is connected to both PCs, we have to take a close look at each Ethernet port. A light (usually green or amber) indicates that the physical connectivity has been established between the two PCs. Then we can try unplugging the cable from one PC, then reconnecting it to verify that the light goes off then back on.

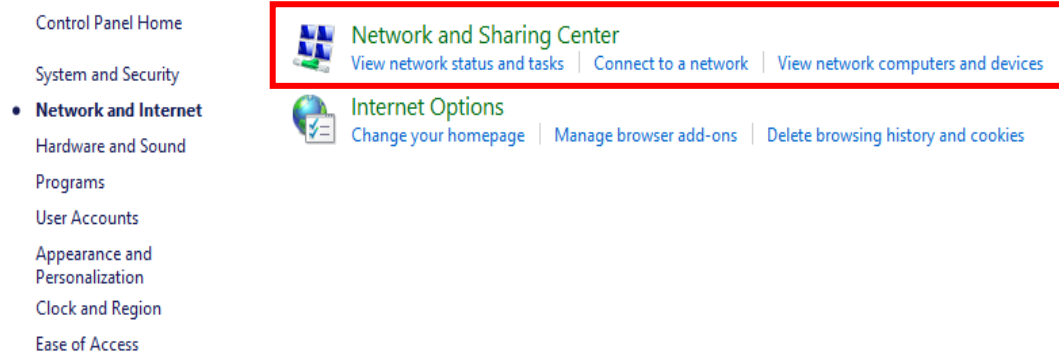
(b) Going to Control Panel, double-clicking the Network Connections icon, and confirm that the local area connection is established. The following figure shows an active local area connection. If physical connectivity problem exist, we will see a red (x) over the local Area Connection icon with the words Network Cable Unplugged.

Step 5: Open Control Panel.

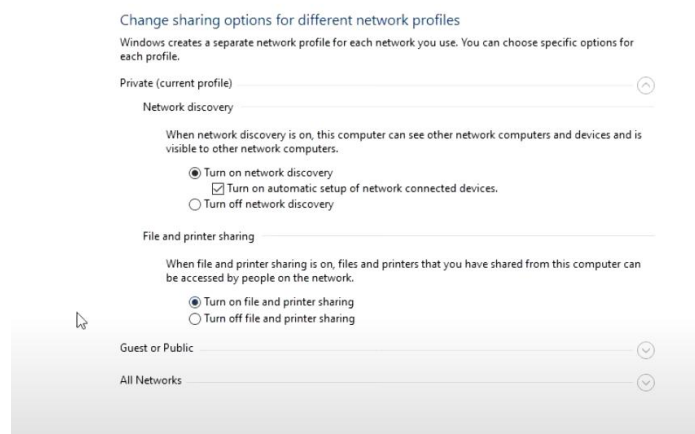
(a) Navigated into the control panel and clicked on the Network and Internet section.



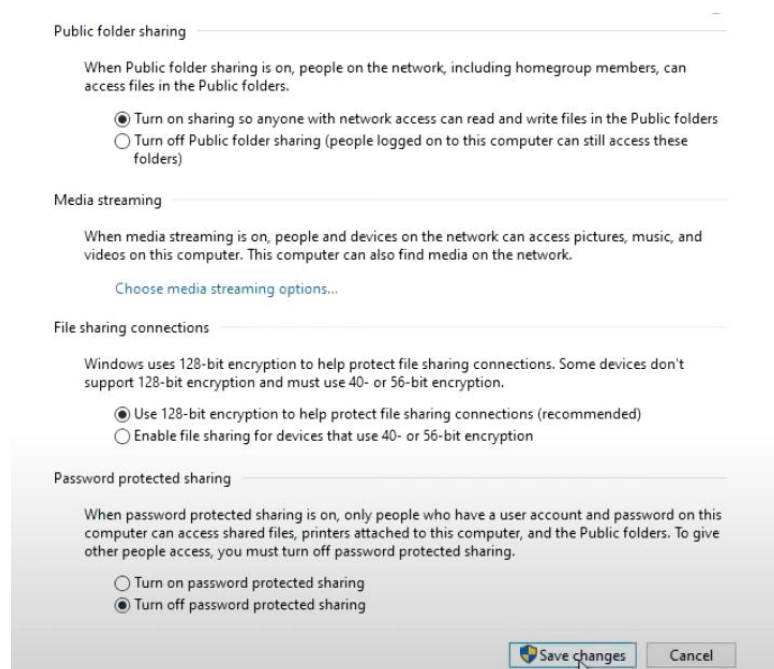
(b) Navigated into the Network and sharing section.



(c) Choose Home and Work / Public. There are many settings that need to be changed in both of the options.

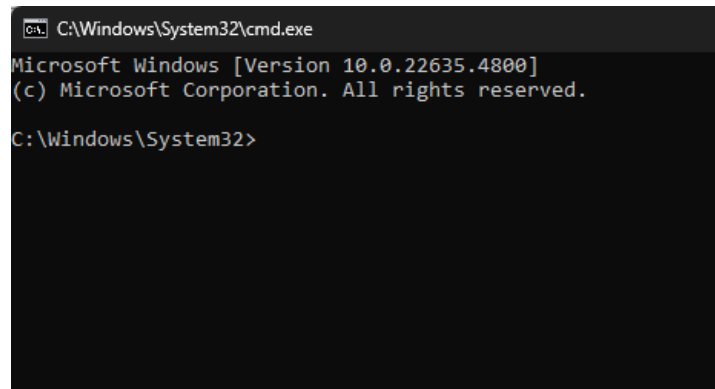


(d) There are going to be many options, the ones we need for the sharing to work are pretty common sense like making sure that our device is allowed to be discovered. And turn off password protected sharing.



Step 6: Navigate to the Desktop.

(a) Opened command prompt and then use the command `<cd desktop>` to change into the desktop directory. This step is simply for convenience so that it is easier to find the folder where we are going to be working with.



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22635.4800]
(c) Microsoft Corporation. All rights reserved.
C:\Windows\System32>
```

(b) Create Folder: Using command "`md folder name`". Making sure that it is visible on our desktop. The command `md` allows us to create a new folder. After typing `md` we will press space and type the name of the folder we want to create. If the folder has more than one word in the folder name we have to make sure that we are putting the name in quotation marks.

Step 7: Find the folder that was shared.

(a) Once we clicked on the device, we will find all the files that were shared from it. We can tell that the folder is shared over the network because it has the green crossroads looking thing under its name.

Name	Date modified	Type	Size
Default	11/29/2020 1:07 AM	File folder	
Public	9/27/2020 9:38 PM	File folder	
smchi	11/28/2020 11:52 AM	File folder	

Discussion:

- This lab provided a hands-on experience in establishing a basic peer-to-peer network using a crossover cable. We learned that successful network communication hinges on several key factors. Firstly, proper physical connectivity is paramount. Using a crossover cable allowed for direct communication between the two PCs, eliminating the need for a central device like a router or switch. We verified the physical connection by observing the LED indicators on the network interface cards.
- Secondly, correct network configuration is crucial. Assigning appropriate IP addresses to each PC and ensuring unique computer names enabled the devices to identify and communicate with each other on the network.
- Finally, enabling sharing settings allowed us to access resources, such as files and printers, on one PC from the other. We explored the Windows Control Panel to configure sharing settings and successfully shared a folder between the two machines.
- Troubleshooting played a vital role throughout the lab. We encountered minor issues with network connectivity and learned to troubleshoot them by checking cable connections, verifying network settings, and identifying potential conflicts.

Caution:

- Security risks, such as unauthorized access, are a concern.
- Performance limitations might be encountered.
- Scalability issues could arise with a large number of devices.

Conclusion:

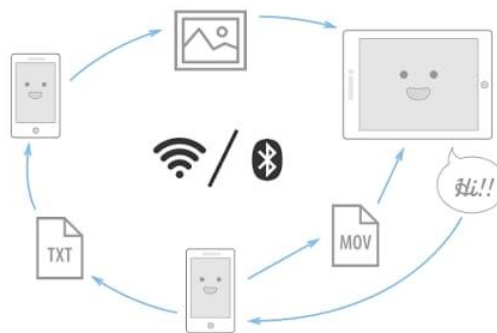
This lab served as a valuable foundation for understanding fundamental network concepts. It highlighted the crucial role of network settings in establishing successful communication and enabling resource sharing between devices. By configuring IP addresses, computer names, and sharing settings, we were able to connect two computers and share resources. The knowledge gained from this experiment, such as troubleshooting techniques and the importance of proper configuration, can be applied to more complex network scenarios, providing a strong foundation for further exploration in network technology.

Experiment No: 04

Experiment Name: Peer-to-Multi-Peer File Sharing Using Layer 2 Switch.

Objective:

1. To establish a peer-to-multi-peer file-sharing network using Windows operating systems and a Layer 2 switch.
2. To configure file-sharing settings on a Windows PC to allow multiple peers to access shared resources.
3. To understand the role of a Layer 2 switch in facilitating data transmission using MAC addresses.
4. To analyze the behavior of the network during simultaneous multi-peer access.



Multi-peer Connectivity at a glance

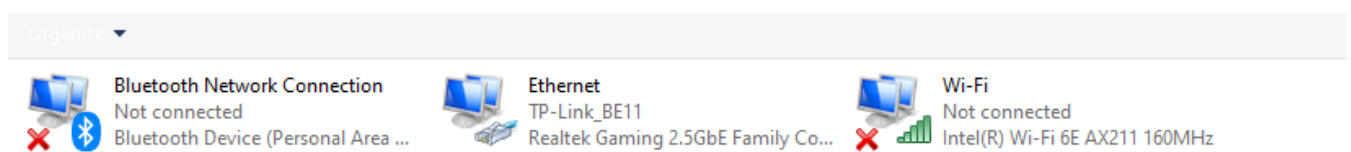
Resources Required:

1. Layer 2 switch
2. Ethernet cables (minimum 3)
3. PCs or laptops (minimum 3) with Windows operating system
4. File-sharing setup via Shared Folders
5. Network configuration tools

Procedure:

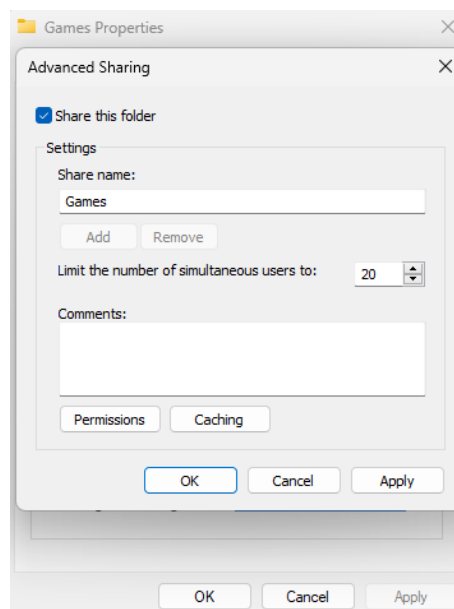
Step 1: Network Setup

- (a) Connect all PCs to the Layer 2 switch using Ethernet cables.
- (b) Ensure that the switch is powered on and functioning properly.
- (c) Assign IP addresses to each PC within the same subnet.
 - Go to Control Panel > Network and Sharing Center > Change Adapter Settings > IPv4 Settings.
 - Set each PC with a unique IP address.
- (d) Verify connectivity between devices using the ping command.



Step 2: Configuring File Sharing on the Server

- (a) Select one PC as the server to share files.
- (b) Create a shared folder:
 - Right-click on the desired folder and select Properties > Sharing > Advanced Sharing.
 - Enable sharing and set appropriate permissions for network users.
- (c) Note the shared folder's network path.
- (d) Test access to the shared folder from another PC by entering the network path in File Explorer.



Step 3: Accessing Files from Peer PCs

- (a) On the client PCs, open File Explorer and navigate to the shared folder using the network path.
- (b) Perform file operations to verify read/write permissions.



Step 4: Testing Multi-Peer Access Simultaneously

- (a) Access the shared folder from multiple client PCs simultaneously.
- (b) Perform concurrent file operations from all clients.
- (c) Observe and note the network's performance and file-sharing behavior.

Name	Date modified	Type	Size
Default	11/29/2020 1:07 AM	File folder	
Public	9/27/2020 9:38 PM	File folder	
smchi	11/28/2020 11:52 AM	File folder	

Step 5: Monitoring and Analyzing Traffic (Optional)

- (a) Use a network monitoring tool to capture and analyze traffic between the peers.
- (b) Filter packets for file-sharing protocols

Discussion:

1. Role of the Layer 2 Switch:

- The Layer 2 switch facilitates peer-to-peer communication by forwarding Ethernet frames based on MAC addresses.
- It ensures efficient and low-latency data transfer without involving IP routing.

2. File Sharing Mechanism:

- Windows Shared Folders utilize the SMB protocol to enable file sharing over a local network.
- The server PC acts as a central node, allowing clients to access shared resources securely.

3. Multi-Peer Access Behavior:

- During simultaneous access, the Layer 2 switch effectively handles traffic by managing multiple MAC addresses in its table.
- The network's efficiency depends on factors such as bandwidth, file size, and the number of simultaneous connections.

4. Challenges Faced:

- Problems with IP configuration were resolved by ensuring all devices were in the same subnet.
- Incorrect file-sharing permissions were adjusted to allow read/write access.
- Minimal delays were observed when multiple peers accessed large files concurrently, highlighting the importance of bandwidth management.

Caution:

- Arrange all devices required for the experiment and ensure they are compatible.
- Manually connect devices to the same network (if required) or enable Bluetooth/Wi-Fi as per the experiment's needs.
- Initiate peer discovery manually on each device and verify that peers are visible to one another.
- Send and receive sample data manually to ensure the communication setup is functional.

Conclusion:

The experiment demonstrated peer-to-multi-peer file sharing on a Windows-based network using a Layer 2 switch. It showcased seamless resource sharing and the efficiency of Layer 2 switches in managing data transmission. Observations highlighted the role of file-sharing protocols, network bandwidth, and proper configuration in ensuring smooth operation.

Experiment No: 05

Experiment Name: Configure a Wireless Router.

Objective:

The objective of this experiment is to:

1. Understand the basic steps to configure a wireless router on a Windows system.
2. Learn how to set up a secure wireless network for home or office use.
3. Gain practical experience in configuring SSID, network security, and IP settings.
4. Ensure proper communication between devices on the network

Resources Required:

1. A wireless router.
2. Windows laptop/PC with an Ethernet port or wireless capability.
3. Ethernet cable (if required for initial setup).
4. Default router login credentials (username and password, typically available on the router or its manual).
5. Internet connection (optional but recommended).

Procedure:**Step 1: Connect to the Router**

- (a) Plug the router into a power source and turn it on.
- (b) Connect your Windows PC to the router using an Ethernet cable or via Wi-Fi (SSID and password are usually printed on the router).
- (c) Ensure the connection is established by checking the network icon in the system tray.



Step 2: Access the Router's Configuration Page

- Open a web browser on your PC.
- Enter the router's IP address into the address bar (commonly 192.168.0.1 or 192.168.1.1).
- Log in using the default username and password.

 Username

 Password

Login

Step 3: Configure Basic Wireless Settings

- Navigate to the "Wireless" or "Wi-Fi Settings" section.
- Set a unique SSID (network name).
- Choose the appropriate wireless mode.

Status

Quick Setup

Network

Wireless

- Basic Settings

- WPS

- Wireless Security

- Wireless MAC Filtering

- Wireless Advanced

- Wireless Statistics

Guest Network

DHCP

Forwarding

Security

Parental Controls

Access Control

Advanced Routing

Bandwidth Control

IP & MAC Binding

Dynamic DNS

IPv6

System Tools

Logout

Wireless Settings

Wireless: ☒ Enable ☐ Disable

Wireless Network Name: (Also called SSID)

Mode:

Channel:

Channel Width:

☒ Enable SSID Broadcast

☐ Enable WDS

Save

Step 4: Configure Network Security

- Navigate to the "Security" or "Wireless Security" tab.
- Select "WPA2-PSK" as the security mode (recommended for better security).
- Set a strong password (minimum 8 characters, a mix of letters, numbers, and symbols).

Status

Quick Setup

Network

Wireless

- Basic Settings
- WPS
- Wireless Security
- Wireless MAC Filtering
- Wireless Advanced
- Wireless Statistics

Guest Network

DHCP

Forwarding

Security

Parental Controls

Access Control

Advanced Routing

Bandwidth Control

IP & MAC Binding

Dynamic DNS

IPv6

System Tools

Logout

Wireless Security Settings

For network security, it is strongly recommended to enable wireless security and select WPA2-PSK AES encryption.

☒ Disable Wireless Security

☐ WPA/WPA2 - Personal(Recommended)

Authentication Type:

Encryption:

Wireless Password:

Group Key Update Period:

☐ WPA/WPA2 - Enterprise

Authentication Type:

Encryption:

RADIUS Server IP:

RADIUS Server Port: (1-65535, 0 stands for default port 1812)

RADIUS Server Password:

Group Key Update Period:

☐ WEP

Authentication Type:

WEP Key Format:

Selected Key: WEP Key Key Type

Key 1:

Key 2:

Key 3:

Key 4:

Step 5: Configure IP Settings (Optional)

- Go to the "LAN Settings" or "Network Settings" section.
- Check if the "DHCP Server" is enabled. This setting automatically assigns IP addresses to connected devices.
- If static IP is required, disable DHCP and manually assign an IP range.

Status

Quick Setup

Network

Wireless

Guest Network

DHCP

- DHCP Settings
- DHCP Clients List
- Address Reservation

Forwarding

Security

Parental Controls

Access Control

Advanced Routing

Bandwidth Control

IP & MAC Binding

Dynamic DNS

IPv6

System Tools

Logout

DHCP Clients List

This page displays information of all DHCP clients on the network.

ID	Client Name	MAC Address	Assigned IP	Lease Time
1	MacBook	74:D4:35:A1:0C:CB	192.168.0.10	00:00:12

Refresh

Step 6: Save and Reboot the Router

- (a) Click "Save" or "Apply" to ensure all settings are updated.
- (b) Reboot the router, either manually or from the admin page.

Step 7: Test the Wireless Network

- (a) Disconnect the Ethernet cable if used.
- (b) Search for the configured SSID on your PC.
- (c) Connect to the network using the newly set password.
- (d) Test the connection by browsing a website or performing a ping test.

Discussion:

Configuring a wireless router involves multiple critical steps to ensure a secure and efficient network. The main points and their sub-points are as follows:

1. Accessing the Router's Admin Panel

- The admin panel is the central hub for all configuration settings.
- Use the default IP address to access the panel.
- Log in with the default credentials provided with the router.

2. Setting Up an SSID

- Ensures the network is easily identifiable for authorized devices.
- Use a unique name to distinguish your network from others.
- Avoid using personal information in the SSID name to enhance privacy.

3. Configuring Network Security

- Protects the network from unauthorized access and potential cyber threats.
- Use WPA2-PSK for strong encryption.
- Set a password that includes letters, numbers, and symbols.

4. Understanding IP Configurations

- Simplifies connectivity by automatically assigning IP addresses to devices.
- Provides fixed IP addresses for specific devices, useful for printers or servers.

5. Maintenance and Updates

- Regular updates optimize performance and security.
- Save configurations to avoid reconfiguring after resets.

Caution:

- Always set a strong and unique password for your Wi-Fi and router admin panel to prevent unauthorized access.
- Regularly check for firmware updates to protect against security vulnerabilities and improve router performance.
- Before making significant changes, back up the router configuration to avoid losing important settings.
- Position the router in a central location to ensure optimal signal coverage throughout the area.
- Do not share your SSID or password publicly, as it increases the risk of unauthorized access.

Conclusion:

This experiment successfully demonstrated how to configure a wireless router using a Windows PC. By following the outlined steps, we set up a secure and efficient wireless network, ensuring optimal performance and protection against unauthorized access. Proper configuration and maintenance practices are essential for sustaining a stable and secure network environment, catering to both personal and professional needs.

Experiment No: 06

Experiment Name: Setup a Network Printer Using Unmanaged Switch.

Objective:

The primary objective of this experiment is to:

1. Understand the process of configuring a network printer in a Windows environment using an unmanaged switch.
2. Establish a basic network connection for a printer and ensure seamless communication between the printer and all connected devices.
3. Learn to troubleshoot common issues encountered while setting up a network printer.

Resources Required:

1. A Windows-based computer or laptop.
2. A network printer with Ethernet capabilities.
3. An unmanaged switch.
4. Ethernet cables (enough to connect all devices).
5. Printer driver/software installation CD or downloaded package.
6. Internet connection (optional for driver installation).

Procedure:

Step 1: Physical Setup of the Printer

- (a) Connect the network printer to the unmanaged switch using an Ethernet cable.
- (b) Connect the Windows computer to the same switch using another Ethernet cable.
- (c) Ensure the unmanaged switch is powered on.
- (d) Turn on the printer and wait until it is ready.
- (e) Verify that both the printer and computer have active network connections.

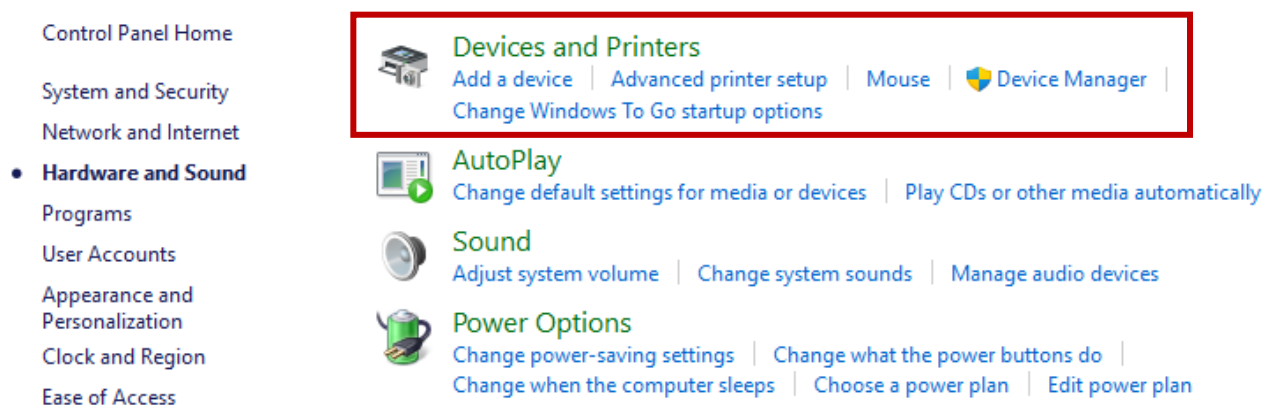


Step 2: Identify Printer's IP Address

- (a) Access the printer's control panel.
- (b) Navigate to the network settings menu.
- (c) Locate and note down the printer's IP address.

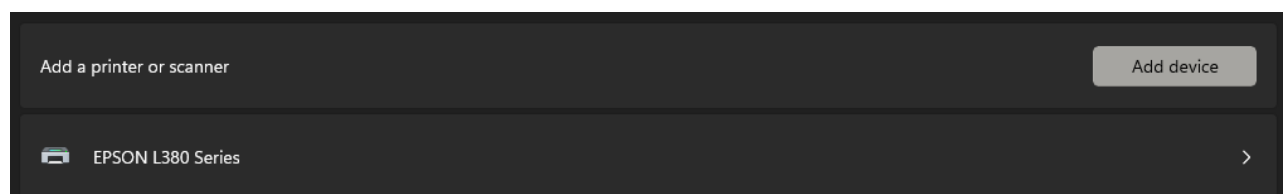
Step 3: Add Printer on Windows

- (a) On your Windows computer, open the Control Panel.
- (b) Navigate to Devices and Printers > Add a Printer.
- (c) Select The printer that I want isn't listed.
- (d) Choose Add a printer using a TCP/IP address or hostname and click Next.
- (e) Enter the printer's IP address noted in Step 2 and click Next.
- (f) Windows will attempt to detect the printer. Install the driver if prompted.



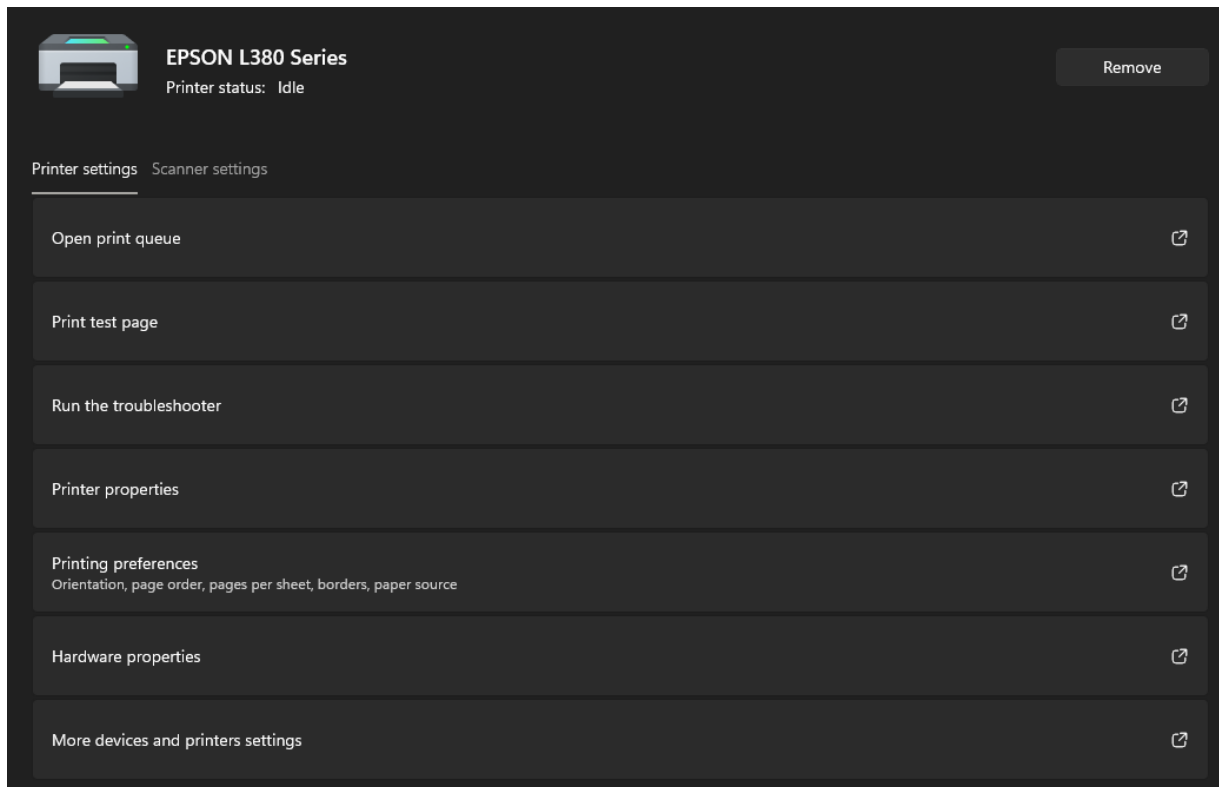
Step 4: Install Printer Drivers

- (a) If Windows cannot find the driver, manually install it using the printer's driver CD or a downloaded package.
- (b) Follow the on-screen instructions to complete the installation process.



Step 5: Configure Printer Settings

- (a) Open Devices and Printers in the Control Panel.
- (b) Right-click on the newly added printer and select Printer Properties.
- (c) Test the printer by sending a test page: Navigate to the General tab and click Print Test Page.



Discussion:

1. Importance of an Unmanaged Switch in Network Printing

(a) Simplicity of Connection

- An unmanaged switch provides an easy, plug-and-play solution for connecting multiple devices.
- It requires no configuration and ensures all devices can communicate seamlessly.

(b) Scalability for Small Networks

- Using an unmanaged switch is ideal for small-scale setups like home or small office networks.
- Multiple devices can connect to the printer without requiring advanced network management skills.

2. Challenges During Setup

(a) Identifying the Printer's IP Address

- Some printers may have complex menus, making it tricky to locate the IP address.
- Having access to the printer's manual can simplify this step.

(b) Driver Compatibility

- Older printers may not have drivers compatible with newer versions of Windows.
- Downloading the latest drivers from the manufacturer's website is often necessary.

3. Ensuring a Reliable Network Connection

(a) Proper Cable Connections

- Loose or damaged Ethernet cables can cause connectivity issues.
- Always use high-quality cables for stable performance.

(b) Printer Configuration

- Ensure the printer's network settings match the intended IP configuration (static or dynamic IP).
- Misconfigured IP settings can lead to communication failures.

Caution:

- Always verify the IP address of the printer before starting the setup process. Using an incorrect IP address can result in failed connections.
- Ensure all Ethernet cables are securely connected and undamaged to avoid network drops.
- Use only official drivers provided by the printer manufacturer to ensure compatibility and stability.
- Avoid using outdated unmanaged switches that may not support modern network speeds effectively.

Conclusion:

Setting up a network printer using an unmanaged switch in a Windows environment is a straightforward process that involves connecting the hardware, configuring the printer's network settings, and installing the necessary drivers on the computer. This experiment highlights the importance of accurate IP address configuration and reliable network connections. By mastering these steps, one can ensure efficient and seamless network printing for small-scale networks.